

CH.JYOTHIRMAI

CH.SC.U4CSE42107

DAA WEEK 5

1.QUICK SORT(FIRST ELEMENT AS PIVOT)

CODE:

```
//CH.SC.U4CSE24107
#include <stdio.h>
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partitionFirst(int a[], int low, int high) {
    int pivot = a[low];
    int i = low + 1;
    int j = high;
    while (i <= j) {
        while (i <= high && a[i] <= pivot)
            i++;
        while (a[j] > pivot)
            j--;

        if (i < j)
            swap(&a[i], &a[j]);
    }
    swap(&a[low], &a[j]);
    return j;
}
void quickSortFirst(int a[], int low, int high) {
    if (low < high) {
        int p = partitionFirst(a, low, high);
        quickSortFirst(a, low, p - 1);
        quickSortFirst(a, p + 1, high);
    }
}
int main() {
    int a[] = {157,110,147,122,111,149,151,141,123,112,117,133};
    int n = 12;
    quickSortFirst(a, 0, n - 1);
    printf("Sorted array (First pivot):\n");
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

OUTPUT:

```
C:\Users\jyoch\OneDrive - Amrita Vishwa Vidyapeetham- Chennai
Sorted array (First pivot):
110 111 112 117 122 123 133 141 147 149 151 157
```

2. QUICK SORT(MIDDLE ELEMENT AS PIVOT)

CODE:

```
//CH.SC.U4CSE24107
#include <stdio.h>
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partitionMiddle(int a[], int low, int high) {
    int mid = (low + high) / 2;
    swap(&a[low], &a[mid]); // middle becomes pivot
    int pivot = a[low];
    int i = low + 1, j = high;
    while (i <= j) {
        while (i <= high && a[i] <= pivot)
            i++;
        while (a[j] > pivot)
            j--;

        if (i < j)
            swap(&a[i], &a[j]);
    }
    swap(&a[low], &a[j]);
    return j;
}
void quickSortMiddle(int a[], int low, int high) {
    if (low < high) {
        int p = partitionMiddle(a, low, high);
        quickSortMiddle(a, low, p - 1);
        quickSortMiddle(a, p + 1, high);
    }
}
int main() {
    int a[] = {157,110,147,122,111,149,151,141,123,112,117,133};
    int n = 12;
    quickSortMiddle(a, 0, n - 1);
    printf("Sorted array (Middle pivot):\n");
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

OUTPUT:

```
C:\Users\jyoch\OneDrive - Amrita Vishwa Vidyapeetham- C
Sorted array (Middle pivot):
110 111 112 117 122 123 133 141 147 149 151 157
C:\Users\jyoch\OneDrive - Amrita Vishwa Vidyapeetham- C
```

3. QUICK SORT(LASTELEMENT AS PIVOT)

CODE:

```
//CH.SC.U4CSE24107
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partitionLast(int a[], int low, int high) {
    int pivot = a[high];
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (a[j] <= pivot) {
            i++;
            swap(&a[i], &a[j]);
        }
    }

    swap(&a[i + 1], &a[high]);
    return i + 1;
}

void quickSortLast(int a[], int low, int high) {
    if (low < high) {
        int p = partitionLast(a, low, high);
        quickSortLast(a, low, p - 1);
        quickSortLast(a, p + 1, high);
    }
}

int main() {
    int a[] = {157,110,147,122,111,149,151,141,123,112,117,133};
    int n = 12;
    quickSortLast(a, 0, n - 1);
    printf("Sorted array (Last pivot):\n");
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}
```

OUTPUT:

```
C:\Users\jyoch\OneDrive - Amrita Vishwa Vidyapeetham- Chennai  
Sorted array (Last pivot):  
110 111 112 117 122 123 133 141 147 149 151 157  
C:\Users\jyoch\OneDrive - Amrita Vishwa Vidyapeetham- Chennai
```